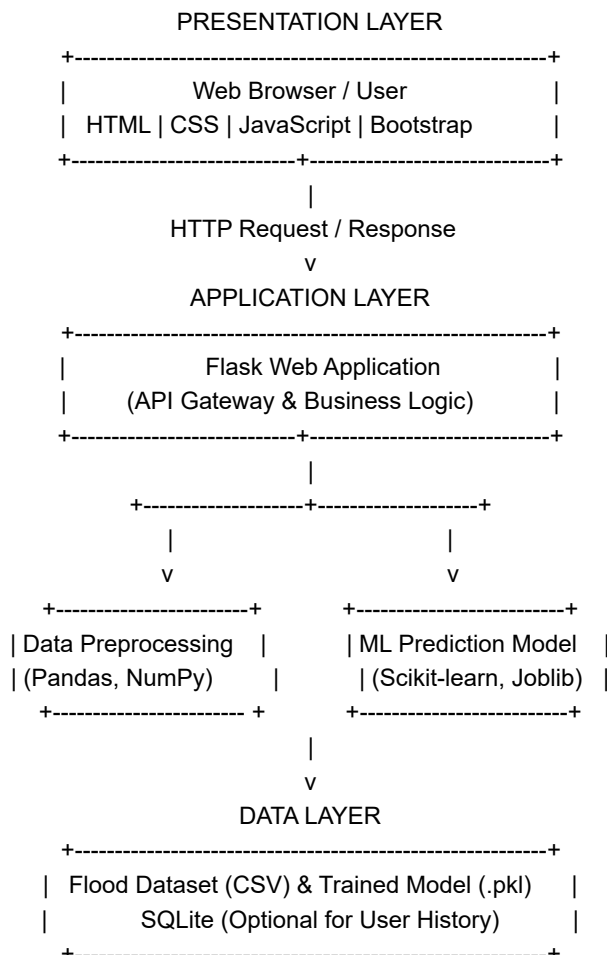


Solution Architecture

Date	04 July 2026
Team ID	
Project Name	Rising Water
Maximum Marks	5 Marks

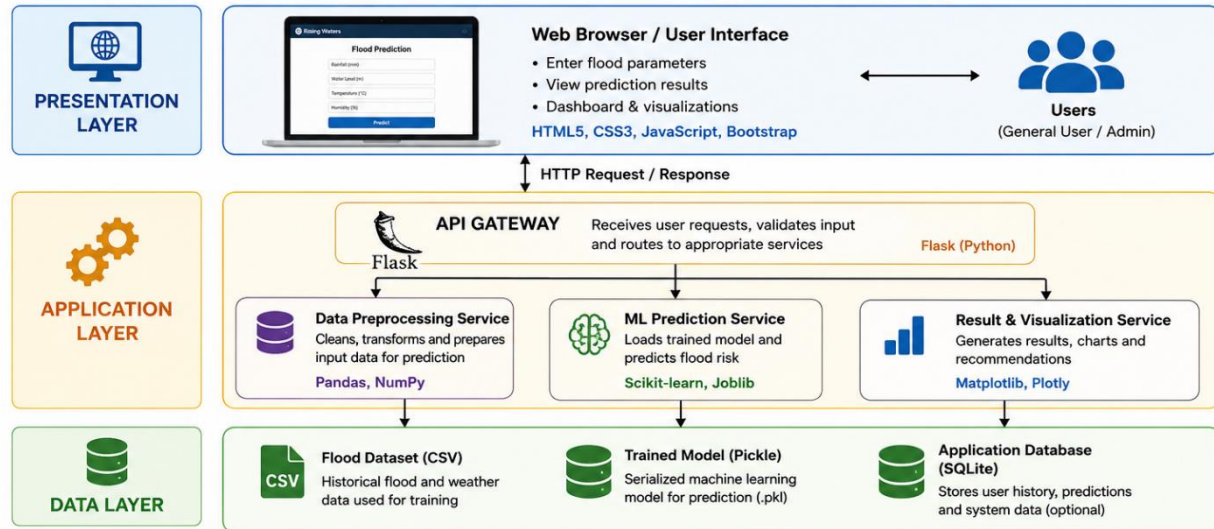
Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



SOLUTION ARCHITECTURE DIAGRAM

Rising Waters – Flood Prediction Using Machine Learning



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where users enter flood parameters and view prediction results.	HTML5, CSS3, JavaScript, Bootstrap
API Gateway	Receives user requests, validates input, and forwards data to the prediction module.	Flask
Microservices	Performs data preprocessing, runs the machine learning model, and generates flood risk predictions.	Python, Pandas, NumPy, Scikit-learn, Joblib
Database / Data Layer	Stores historical flood datasets, trained ML model, and optionally user prediction history.	CSV Dataset, Pickle (.pkl), SQLite (Optional)